

Customer		Date	2026-04-15
Project Title	IT31023A	Code No.	
Project Description			
IC Body	☐ GPL815P04A, OTP size: 32K-byte, SRAM size: 3072-byte, 9COM * 48SEG ☐ GPL815P04B, OTP size: 31K-byte, SRAM size: 3072-byte, 9COM * 48SEG ☒ GPL815P02A, OTP size: 16K-byte, SRAM size: 1536-byte, 8COM * 48SEG		
Release Code File (fill "FFH" for unused area)			
Binary filename	IT31023A.hdb	Binary file checksum	0x001A789A
		User-ROM checksum (computed by G+IDE tool)	0x001A65BA
Important Check			
The OTP body will be referencing the code option during power-on. User needs to select the correct code option in the G+ IDE under the Mask/OTP option field. ☐ Check; by ticking off this box, customer understands that the G+IDE options are fully met and verified.			
Hardware / Software			
The hardware/software programming check list intends to provide some IC specific directions to use Generalplus device. It is the customer's responsibility to check and confirm all the information in the list. No responsibility is assumed by Generalplus if any item is missed by the customer. Make sure the following conditions are met and verified. ● Binary/HDB file is fully tested on OTP chip. ● GPL815PXXB: GP Test Code Area (0x0 ~ 0x3FF 1K-byte), not allowed to be used by user. ● The GPL815P02Ax does not support SPI and I2C interfaces. ● IO : i. GPL815P04Xx : Max. 40 I/O pins (PA[7:0], PB[7:0], PC[7:0], PD[7:0], PE[7:0]), 16 Outputs pins (PF[7:0], PG[7:0]). ii. GPL815P02Ax : Max. 21 I/O pins (PA[7:0], PB[7:0], PC[4:0]), 16 Outputs pins (PF[7:0], PG[7:0]). ● LVR (Code Option) : 2.5V @ system clock = 8.192 MHz 2.2V @ system clock = 4.096 MHz 1.9V @ system clock = 2.048 MHz ● VREG (Code Option) : 2.1V @ system clock ≤ 2.048 MHz 2.5V @ system clock > 2.048MHz ● COM[0:3] or COM[5:8] share PC[7:4] (Code Option) i. COM[0:3] share PC[7:4]: - PC[7:4] set as COM, Max 9 COM - PC[7:4] set as IO, User can't use LCD Function. (Do not support COM[8:4]) ii. COM[5:8] share PC[7:4] : - PC[7:4] set as COM, Max 9 COM - PC[7:4] set as IO, Max 5 COM ● First power on, delay at least 32ms for VREG stable. ● Working Voltage: 2.0V~5.5V. ● Maximum CPU frequency is 2.048MHz or 8.192MHz. (Code Option) ● When system powers on, LCD control port and buffer must be initialized. Otherwise, the data on LCD driver will be randomly generated. ● ADC Power (VDD_ADC) can't over 3.6V. ● System Clock from IOSC, there may be frequency deviations that could lead to unstable UART communication - Users can utilize the Code Option "UART calibration with X32K" to externally connect a 32768 crystal, allowing for system frequency calibration using the 32768 crystal. ● When segment outputs are shared with key-scan output port and PortA is used for key-scan input port (pins), all PortA pins for key-scan input must be configured as pull high state. ● If SPI, I2C, UART interfaces are used: To avoid extra power consumption, be sure to set SPI, I2C, UART I/Os to the proper status before entering low power mode. ● Stack area located at 0x100 – 0x1FF ● PortA or PortB data must be latched before setting wakeup source and before entering sleep mode. That is, the following code must be presented prior to setting wakeup source and entering sleep mode: LDA \$308A ; latch PortA LDA \$3092 ; latch PortB			

☐ Check; by ticking off this box, customer understands that the above conditions are fully met and verified.

General Programming Checklist

The general programming checklist intends to provide some general characteristics about GENERALPLUS devices. It is the customer's responsibility to check all the information in the list. No responsibility is assumed by GENERALPLUS for any non-checked box even this confirmation sheet has been approved by GENERALPLUS. Make sure the following conditions are met and verified:

- CPU stack pointer must be reset after system power-on.
- All RAM must be initialized after power on.
- Do not enable interrupt before initializing RAM.
- Sleep port must be cleared and re-initialized before entering sleep mode.
- The used RAM not over the stack reserved area.
- Correct RAM/ Flash size and start addresses.
- No current drain on I/O in sleep mode.
- No I/O remains floating in sleep mode.
- Non-used I/O ports should be masked off (for input process).
- Example (suppose PortA[3:0] as input and PortA [7:4] are invalid):
LDA PortA ; read I/O port A Data
AND #0FH ; mask off high nibble

☐ Check; by ticking off this box, customer understands that the above conditions are fully met and verified.

Development Tool / Board Version

Programming guide title and version

User's guide title and version

Other documents (if any)

Customer's Signature

簽名指引

Note: Please send/fax this form to GENERALPLUS. GENERALPLUS will return it back with signature.